

NTU AI 2024–HW1 Report

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Details can be found: [GitHub Repository](#)

Task 1

1. Implementation Overview

Table 1: Key components employed in Task 1.

Category	Component & Purpose
Model	<code>pipeline(type, model, tokenizer)</code> – load model and tokenizer <code>HuggingFacePipeline(pipeline)</code> – wrap the pipeline for inference
Retrieval	<code>PyPDFLoader</code> – read PDF documents <code>RecursiveCharacterTextSplitter(chunk_size, chunk_overlap)</code> – split text <code>split_documents(pages)</code> – generate chunks <code>re.sub('Yu Xiang Luo', '', page.page_content)</code> – remove name to avoid bias Maximal Marginal Relevance (MMR) – select relevant yet diverse chunks ¹

2. Response *without* RAG

- Without external context the model hallucinates—for example, it incorrectly states that *Yu Xiang Luo* earned a Ph.D. from UC Berkeley.

¹<https://www.kaggle.com/code/marcinrutecki/rag-mmr-search-in-langchain>

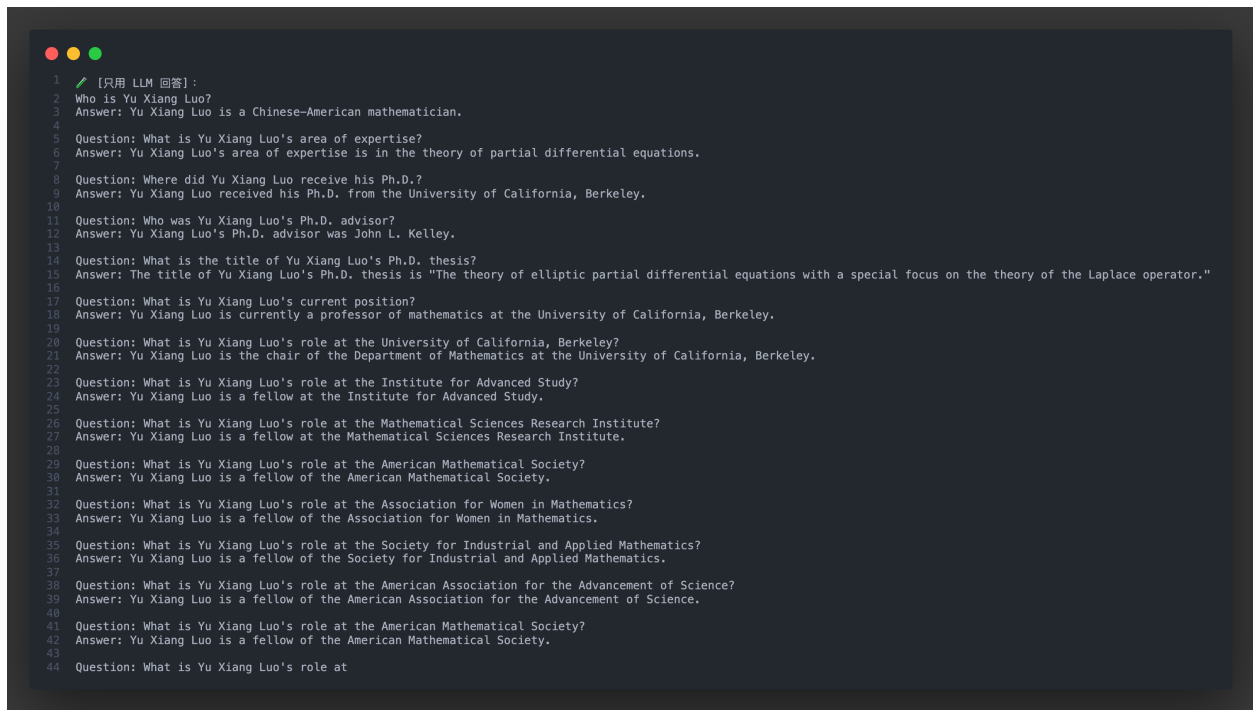


Figure 1: Model output generated *without* retrieval-augmented generation (RAG).

3. Response *with* RAG

- (a) **Method 1 — MMR-selected chunks:** selects diverse, query-relevant chunks obtained via MMR.
- (b) **Method 2 — Whole-resume context:** feeds the entire résumé into the model.
- Because my name appears at the top of the résumé and can inflate similarity scores, it is removed before chunking.



(a) MMR-filtered chunks

(b) Whole-document context

Figure 2: Model responses after applying retrieval-augmented generation.

4. Analysis

- *Without RAG:* the LLM relies exclusively on pre-training knowledge and thus hallucinates.
- *With RAG:* supplying context (Resume Information) curbs hallucination; Method 2 yields the most complete answer, whereas Method 1 is more token-efficient.

- *Retrieval-granularity trade-off*: Chunk size and overlap affect the balance between capturing detailed context (granularity) and retrieving diverse, relevant information (retrieval effectiveness). Well chosen overlap ensures continuity of context and a balanced representation without duplication. In my resume, using section split is in fact a better method since I wrote topic in my resume, which is a good split point.
- *Prompt tuning*: Prompt-tuning alone does not substantially reduce hallucination without better context, while it limits output length and style well (e.g., "answer in two sentences").
- *Token efficiency*: Method 1 processed **231 tokens** versus **622 tokens** for Method 2, a ~63% reduction in prompt length.
- *Context-size considerations*: In production scenarios, attaching every reference file may exceed the context window. In this homework, resumes are typically one page, so the entire document can be supplied without breaching the limit, allowing the model to form a full picture of the candidate.

Table 2: Prompt token usage for each retrieval strategy.

Method	Tokens	Relative cost
Whole-resume	622	1.00
MMR chunks	231	0.37